

E.G.S. PILLAY ENGINEERING COLLEGE (AUTONOMOUS)

Approved by AICTE, New Delhi

(Affiliated to Anna University, Chennai | Re-accredited by NAAC with 'A++' Grade)

Accredited by NBA (B.Tech - IT, B.E-CIVIL, CSE, ECE, EEE, MECH) (Tier-1)

NAGAPATTINAM – 611002



B.TECH - INFORMATION TECHNOLOGY (R-2023)

THIRD YEAR- FIFTH SEMESTER

Course Code	Course Name	Category	L	T	P	C	Max. Marks		
							CA	ES	Total
THEORY COURSES									
2302IT501	Object Oriented Analysis and Design	PCC	3	0	0	3	40	60	100
2302IT502	Web Programming	PCC	3	0	0	3	40	60	100
2302IT503	Mobile Computing	PCC	3	0	0	3	40	60	100
2302IT504	Artificial Intelligence and Machine Learning	PCC	3	0	2	4	50	50	100
	Professional Elective I	PEC	3	0	0	3	40	60	100
	Open Elective I	OEC	3	0	0	3	40	60	100
LABORATORY COURSES									
2302IT551	Web Programming Lab	PCC	0	0	2	1	60	40	100
2304IT551	CASE Tools Lab (Mini Project I)	EEC	0	0	2	1	60	40	100
	Professional Development Course III	EEC	0	0	2	1	60	40	100
2304IT552	Startup Opportunities for IT Engineers	EEC	2	0	0	2	100	-	100
MANDATORY COURSES									
2304GE551	Life skills V	MC	0	0	0	0	100	-	100
Total			20	0	8	24	630	470	1100

2302IT501	OBJECT ORIENTED ANALYSIS AND DESIGN								L	T	P	C
									3	0	0	3
PREREQUISITE:												
Software Engineering and Project Management												
COURSE OBJECTIVES:												
1) To Learn the basics of OO analysis and design skills.												
2) To Learn the UML design diagrams.												
3) To improve the software design with design patterns.												
4) To be exposed to the various testing techniques.												
COURSE OUTCOMES:												
After completion of the course, Student will be able to												
CO1:	Design and implement projects using OO concepts											
CO2:	Use the UML analysis and design diagrams											
CO3:	Apply appropriate design patterns											
CO4:	Create code from design											
CO5:	Compare and contrast various testing techniques											
COs Vs POs MAPPING:												
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
	CO1	2		2	3						3	2
	CO2			3	3							1
	CO3	1		3	3						2	
	CO4		2				3	3			2	
	CO5					2	3	3			1	
COs Vs PSOs MAPPING:												
	COs	PSO1	PSO2	PSO3								
	CO1	1										
	CO2		2									
	CO3			1								
	CO4		1									
	CO5			1								
COURSE CONTENTS:												
UNIT I	UML DIAGRAMS										9 Hours	
Introduction to OOAD – Unified Process - UML diagrams – Use Case – Class Diagrams– Interaction Diagrams – State Diagrams – Activity Diagrams – Package, component and Deployment Diagrams												
UNIT II	DESIGN PATTERNS										9 Hours	
GRASP: Designing objects with responsibilities – Creator – Information expert – Low Coupling – High Cohesion – Controller - Design Patterns – Creational - Structural -Behavioral.												
UNIT III	CASE STUDY										9 Hours	
the Next Gen POS system, Inception -Use case Modeling - Relating Use cases - Elaboration - Domain Models - Finding conceptual classes and description classes – Associations – Attributes – Domain model refinement – Finding conceptual class Hierarchies - Aggregation and Composition												
UNIT IV	APPLYING DESIGN PATTERNS										9 Hours	
System sequence diagrams - Relationship between sequence diagrams and use cases Logical architecture and UML package diagram – Logical architecture refinement - UML class diagrams – UML interaction diagrams - Applying GoF design patterns												
UNIT V	TESTING										9 Hours	

Object Oriented Methodologies — Software Quality Assurance — Mapping design to code – – methodology – Survey – Rum Baugh, Booch, Jacobson Methods Testing: Issues in OO Testing – Class Testing – OO Integration Testing –GUI Testing – OO System Testing- Case Study : Vacation Tracking System
TOTAL: 45 HOURS
REFERENCES:
1. Craig Larman, "Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development", Third Edition, Pearson Education, 2016.
2. Simon Bennett, Steve Mc Robb and Ray Farmer, "Object Oriented Systems Analysis and Design Using UML", Fourth Edition, Mc-Graw Hill Education, 2012.
3. Erich Gamma, a n d Richard Helm, Ralph Johnson, John Vlissides, "Design patterns: Elements of Reusable Object-Oriented Software", Addison-Wesley, 2012.
4. Martin Fowler, "UML Distilled: A Brief Guide to the Standard Object Modeling Language", Third edition, Addison Wesley, 2013.
5. Paul C. Jorgensen, "Software Testing:- A Craftsman's Approach", Third Edition, Auerbach Publications, Taylor and Francis Group, 2010.
6. http://nptel.ac.in/

2302IT502	WEB PROGRAMMING	L	T	P	C
		3	0	0	3

PREREQUISITE: Programming in Java Programming, Database Management Systems

COURSE OBJECTIVES:

1. Understand the scripting languages XHTML, JavaScript and PHP
2. Familiar with the different server technologies
3. Gain knowledge in the concepts of web services.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Demonstrate the technologies used to create web pages
CO2:	Design dynamic and interactive web pages by embedding Java Script in XHTML
CO3:	Develop server side programming and apply database concepts to solve real time application
CO4:	Build web applications using PHP and to represent data in XML format
CO5:	Use AJAX and web services to develop interactive web applications

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1		3						2
CO2	3	3	3		3						3
CO3	3	3	3	2	3					2	3
CO4	3	3	3	2	3					2	3
CO5	3	3	3	2	3					2	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	2	1
CO2	2	3	3
CO3	3	3	3
CO4	3	3	3
CO5	3	3	3

COURSE CONTENTS:

UNIT I	WEB DESIGNING	9 Hours
Markup languages – Introduction to XHTML-Editing XHTML-Headings- Linking –Tables-Images-Forms- Internal linking – Frames – Lists Cascading Style Sheets (CSS): Features-Style rule cascading and inheritance - Text properties –CSS box model		
UNIT II	CLIENT SIDE PROGRAMMING	9 hours
Introduction to java script –Control statements I– Control statements II - Functions- Objects – Arrays- Introduction to JSON.		
UNIT III	SERVER SIDE PROGRAMMING	9 Hours
Servlets: Java Servlet Architecture- Servlet Life Cycle- Form GET and POST actions- Session Handling- Understanding Cookies- Installing and Configuring Apache Tomcat Web Server DATABASE CONNECTIVITY: JDBC perspectives, JDBC program example — JSP: Understanding Java Server Pages-JSP Standard Tag Library (JSTL)-Creating HTML forms by embedding JSP code		
UNIT IV	PHP & XML	11 hours
An introduction to PHP: PHP- Using PHP- Variables- Program control- Built-in functions- Form Validation- Regular Expressions — File handling — Cookies — Connecting to Database. XML: Basic XML- DTD - XML Schema, XML Parsers and Validation, XSLT Transformation		
UNIT V	INTRODUCTION AJAX AND WEB SERVICES	7 Hours
AJAX: Ajax Client Server Architecture-XML Http Request Response Object Web Services: Introduction- Java web services Basics — Creating, Publishing, Testing and Describing a Web services (WSDL)-Consuming a web service-SOAP.		
TOTAL: 45 HOURS		
REFERENCES:		
1. Paul Deitel, Harvey Deitel, Abbey Deitel, “Internet & World Wide Web - How to Program”, 5 th edition, Pearson Education, 2012.		
2. Robin Nixon, "Learning PHP, MySQL, JavaScript, CSS & HTML5" Sixth Edition, O'Reilly publishers, 2022.		
3. Jeffrey C and Jackson, “Web Technologies A Computer Science Perspective”, Pearson Education, 2011.		
4. Steven Holzener , “PHP – The Complete Reference”, 1st Edition, Mc-Graw Hill, 2017		
5. Gopalan N.P. and Akilandeswari J., “Web Technology”, Prentice Hall of India, 2011		
6. UttamK.Roy, “Web Technologies”, Oxford University Press, 2011		
7. Learning Resources https://www.freecodecamp.org/ https://www.w3schools.com/ https://www.codecademy.com/		

2302IT503	MOBILE COMPUTING	L	T	P	C
		3	0	0	3
PREREQUISITE:					
Principles of Communication, Computer Networks					
COURSE OBJECTIVES:					
1) To understand the basic concepts of mobile computing. 2) To learn the basics of mobile telecommunication system. 3) To be familiar with the network layer protocols and Ad-Hoc networks. 4) To know the basis of transport and application layer protocols. 5) To gain knowledge about different mobile platforms and application development.					

After completion of the course, Student will be able to	
CO1:	Explain the concepts of mobile computing schemes and applications
CO2:	Illustrate the generations of telecommunication systems in wireless networks
CO3:	Determine the functionality of MAC, network layer and Identify a routing protocol for a given Ad hoc network
CO4:	Explain the functionality of Transport and Application layers
CO5:	Develop a mobile application using android/blackberry/Ios/Windows SDK

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2		-	-				2		2
CO2	3	2	-	-	-				2		3
CO3	3	3	2	2	-						2
CO4	3	2	-	2	-						2
CO5	3	3	3	3	3	2		2	2	3	3

	COs	PSO1	PSO2	PSO3
	CO1	2	2	1
	CO2	3	1	1
	CO3	3	1	2
	CO4	3	1	2
	CO5	2	3	3

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1.	Prasant Kumar Pattnaik, Rajib Mall, “Fundamentals of Mobile Computing”, PHI Learning Pvt. Ltd, New Delhi – 2016.
2.	Jochen H. Schller, “Mobile Communications”, Second Edition, Pearson Education, New Delhi, 2014.

3.	Dharma Prakash Agarval, Qing and An Zeng, "Introduction to Wireless and Mobile systems", Thomson Asia Pvt Ltd, 2013.
4.	Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, "Principles of Mobile Computing", Springer, 2012.
5.	William.C.Y.Lee,"Mobile Cellular Telecommunications-Analog and Digital Systems", Second Edition, Tata Mc Graw Hill Edition ,2012.
6.	C.K.Toh, "AdHoc Mobile Wireless Networks", First Edition, Pearson Education, 2012.
7.	http://nptel.ac.in/

2302IT504	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING	L	T	P	C							
		3	0	2	4							
PREREQUISITE:												
	1. Datamining											
	2. Programming for problem solving											
COURSE OBJECTIVES:												
	1. Learn about the fundamentals of AI and ML.											
	2. Learn about the probabilistic reasoning.											
	3. Understand the Supervised learning with various techniques.											
	4. Study the extraction of k-means ,KNN, Gaussian models and Expectation maximization											
	5. Learn about Machine Learning based on various applications											
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Demonstrate the core concepts of Artificial Intelligence and Machine Learning with architecture.											
CO2:	Apply AI problem-Solving Techniques to implement common AI strategies using algorithms.											
CO3:	Implement Machine Learning Models (Supervised and Unsupervised).											
CO4:	Analyse and combining multiple learners using different techniques.											
CO5:	Understand the application of AI in areas like performance measurement and Image recognition, etc											
COs Vs POs MAPPING:												
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
	CO1	3	3	3	-	3	-	-	-	-	-	3
	CO2	3	2	2	-	2	-	-	-	-	-	2
	CO3	3	3	3	-	3	-	-	-	-	-	3
	CO4	3	3	3	-	3	-	-	-	-	-	3
	CO5	3	2	2	-	2	-	-	-	-	-	2
COs Vs PSOs MAPPING:												

<table><tr><th>COs</th><th>PSO1</th><th>PSO2</th><th>PSO3</th></tr><tr><td>CO1</td><td>3</td><td>3</td><td>2</td></tr><tr><td>CO2</td><td>3</td><td>2</td><td>2</td></tr><tr><td>CO3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>CO4</td><td>3</td><td>3</td><td>2</td></tr><tr><td>CO5</td><td>3</td><td>2</td><td>2</td></tr></table>				COs	PSO1	PSO2	PSO3	CO1	3	3	2	CO2	3	2	2	CO3	3	3	2	CO4	3	3	2	CO5	3	2	2
COs	PSO1	PSO2	PSO3																								
CO1	3	3	2																								
CO2	3	2	2																								
CO3	3	3	2																								
CO4	3	3	2																								
CO5	3	2	2																								
COURSE CONTENTS:																											
MODULE I	INTRODUCTION TO DATA WAREHOUSING																										
9 Hours																											
Introduction to AI – AI applications – problem solving agents – search algorithms – uninformed search strategies – Heuristic search strategies – Local search and optimization problems – advanced search – constraint satisfaction problems (CSP).																											
MODULE II	PROBABILISTIC REASONING																										
9 Hours																											
Acting under uncertainty – Bayesian inference - naïve Bayesian models, Probabilistic reasoning - Bayesian networks – exact inference in BN – approximate inference in BN – casual networks.																											
MODULE III	SUPERVISED LEARNING																										
9 Hours																											
Introduction to machine learning – Linear Regression Models: Least squares, single & multiple variables, Bayesian linear regression, gradient descent, Linear Classification Models: Discriminant function – Probabilistic discriminative model - Logistic regression, Probabilistic generative model – Naive Bayes, Maximum margin classifier – Support vector machine, Decision Tree, Random forests																											
MODULE IV	ENSEMBLE TECHNIQUES AND UNSUPERVISED LEARNING																										
9 Hours																											
Combining multiple learners: Model combinations schemes, Voting, Ensemble Learning – bagging, Boosting, stacking, Unsupervised learning: k – means, Instance based Learning: KNN, Gaussian mixture models and Expectation maximization.																											
MODULE V	MACHINE LEARNING IN PRACTICE AND APPLICATIONS																										
9 Hours																											
Perceptron - Multilayer perceptron, activation functions, network training – gradient descent optimization – stochastic gradient descent, error back propagation, from shallow networks to deep networks –Unit saturation (aka the vanishing gradient problem) – ReLU, hyper parameter tuning, batch normalization, regularization, dropout.																											
TOTAL: 45 HOURS																											
PRACTICAL EXERCISES:																											
1. Implementation of Uninformed search algorithms (BFS, DFS) 2. Implementation of Informed search algorithms (A*, memory-bounded A*) 3. Write a program to implement the naïve Bayesian classifier for credit card analysis and compute the accuracy with a few test data sets 4. Build Regression models 5. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample. 6. Build SVM models 7. Implement Ensembling techniques 8. Implement clustering algorithms 9. Implement EM for Bayesian networks																											

10. Build deep learning NN models

List of Equipments:(30 Students per Batch)

Tools: Python, Numpy, Scipy, Matplotlib, Pandas, statmodels, seaborn, plotly, bokeh

Note: Example data sets like: UCI, Iris, Pima Indians Diabetes etc

TOTAL: 30 HOURS

REFERENCES:

1. Stuart Russell and Peter Norvig, “Artificial Intelligence – A modern Approach”, Fourth Edition , Pearson Education, 2021.
2. Ethm Alpaydin, “ Introduction to Machine Learning “, MIT Press , Fourth Edition, 2020
3. Ameet V Joshi, —Machine Learning and Artificial Intelligence, Springer Publications, 2020.
4. Dan W. Patterson, “Introduction to Artificial Intelligence and Expert Systems”, Pearson Education, 2007
5. Education, 2007
6. <http://nptel.ac.in/>
7. <http://www.simpilearn.com/>
8. <http://towardsdatascience.com/>

2302IT551	WEB PROGRAMMING LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE: Programming in Java Programming, Database Management Systems

COURSE OBJECTIVES:

1. To develop an ability to design and implement static and dynamic website
2. Use appropriate client-side or Server-side applications
3. Understand, analyze and create XML documents and XML Schema

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Create web pages using HTML and Cascading Styles sheets
CO2:	Create dynamic web pages using JavaScript
CO3:	Develop Servlet and JSP applications to implement Session management and Data base Connectivity
CO4:	Build web applications using PHP
CO5:	Design and validate XML documents.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	3		3						1
CO2	3	2	3		3						2
CO3	3	3	3	2	3					2	3
CO4	3	3	3	2	3					2	3
CO5	3	2	2		3						2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	2	2
CO2	2	3	3
CO3	3	3	3
CO4	3	3	3
CO5	3	2	3

List of Experiments**Experiment 1 (HTML Page)**

- (a) Design the following static web pages required for an online book store web site.
- (b) HOME PAGE: The static home page must contain three frames.
- (c) LOGIN PAGE
- (d) CATALOGUE PAGE: The catalogue page should contain the details of all the books available in the web site in a table.
- (e) REGISTRATION PAGE

Experiment 2 (CSS)

- (a) Develop and demonstrate the usage of inline, internal and external style sheet using CSS

Experiment 3 (JavaScript)

Write JavaScript to validate the following fields of the Registration page.

- (a) First Name (Name should contains alphabets and the length should not be less than 6 characters).
- (b) Password (Password should not be less than 6 characters length).
- (c) E-mail id (should not contain any invalid and must follow the standard pattern name@domain.com)
- (d) Mobile Number (Phone number should contain 10 digits only).
- (e) Last Name and Address (should not be Empty).

Experiment 4 (Installing Servers)

- (a) Install TOMCAT web server and APACHE. While installation assign port number 4040 to TOMCAT and 8080 to APACHE. Make sure that these ports are available i.e., no other process is using this port.
- (b) Access the following static web pages - an online book store web site using these servers
- (c) Access the pages by using the urls : <http://localhost:4040/rama/books.html> (for tomcat)
<http://localhost:8080/books.html> (for Apache)

Experiment 5 (Installing Databases)

- (a) Install a database (Mysql or Oracle).
- (b) Create a table which should contain at least the following fields: name, password, email-id, phone number (these should hold the data from the registration form).
- (c) Practice 'JDBC' connectivity.
- (d) Write a java program/servlet/JSP to connect to that database and extract data from the tables and display them.
- (e) Experiment with various SQL queries. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.

Experiment 6, 7 & 8 (Servlet, JSP and PHP)

- (a) A web application that takes a name as input and on submit it shows a hello <name> page where name is taken from the request.
- (b) It shows the start time at the right top corner of the page and provides a logout button.
- (c) On clicking this button, it should show a logout page with Thank You <name > message with the duration of usage (hint: Use session to store name and time).

Experiment 9 (XML with DOM)

- (a) Create an XML document that contains 10 users information.
- (b) Write a Java Program, which takes User Id as input and returns the user details by taking the user information from XML document using DOM parser or SAX parser.

TOTAL: 30 HOURS	
REFERENCES:	
1.	Paul Deitel, Harvey Deitel, Abbey Deitel, “Internet & World Wide Web - How to Program”, 5th edition, Pearson Education, 2012.
2.	Robin Nixon, "Learning PHP, MySQL, JavaScript, CSS & HTML5" Third Edition, O'Reilly publishers, 2014.
3.	Jeffrey C and Jackson, “Web Technologies A Computer Science Perspective”, Pearson Education, 2011.
4.	Steven Holzener , “PHP – The Complete Reference”, 1st Edition, Mc-Graw Hill, 2017
5.	Gopalan N.P. and Akilandeswari J., “Web Technology”, Prentice Hall of India, 2011
6.	UttamK.Roy, “Web Technologies”, Oxford University Press, 2011
7.	Learning Resources https://www.freecodecamp.org/ https://www.w3schools.com/ https://www.codecademy.com/

2304IT551	CASE TOOLS (MINI PROJECT 1)	L	T	P	C
		0	0	2	1

PREREQUISITE:

Software Engineering and Project Management

COURSE OBJECTIVES:

1.	Understand the concept of modeling and mechanism involved in UML.
2.	Learn the classes and different types of relationships in classes, objects and terms related to diagrams.
3.	Examine fundamental object-oriented analysis and design techniques.
4.	Apply design patterns for viewing a system as a set of procedures.
5.	Prepare case studies for analyzing modeling techniques.

COURSE OUTCOMES:

After completion of the course, Student will be able to

CO1:	Design and implement projects using OO concepts.
CO2:	Use the UML analysis and design diagrams.
CO3:	Apply appropriate design patterns.
CO4:	Create code from design.
CO5:	Compare and contrast various testing techniques

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	3			2	2	2	3
CO2	2	3	3	2	3				3		2
CO3	3	3	3	1	2						3
CO4	3	2	3	2	3						2
CO5	3	3	1	3	2						3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	3	3
CO2	1	2	3
CO3	1	3	3
CO4	1	3	3
CO5	1	2	3

TO DEVELOP A MINI-PROJECT USING FOLLOWING PROBLEM STATEMENTS**Identify Use Cases and develop the Use Case model.**

- Identify the conceptual classes and develop a domain model with UML Class diagram.
- Using the identified scenarios, find the interaction between objects and represent them using UML Sequence diagrams.
- Draw relevant state charts and activity diagrams.
- Test the software system for all the scenarios identified as per the usecase diagram
- Identify the User Interface, Domain objects, and Technical services. Draw the partial layered, logical architecture diagram with UML package diagram notation.
- Implement the modified system and test it for various scenarios.
- Develop and test the Technical services layer.
- Develop and test the Domain objects layer.
- Develop and test the User interface layer

Domains for Mini Project

- Online Purchase System
- Library Management System
- E-Ticketing
- Quiz System
- Student Mark Analyzing System
- E-Mail Client System
- Telephone Phone Dialing
- Point of Sale
- Software personnel management system
- Airline/Railway reservation system

Requirements: Argo UML or Eclipse IDE or Rational Suite or Visual Paradigm or equivalent**TOTAL: 30 HOURS****REFERENCES :**

1. Bernd Bruegge and Allen H. Dutoit, "Object-Oriented Software Engineering: Using UML, Patterns and Java", Third Edition, Pearson Education, 2009.
2. Roger S. Pressman, Object-Oriented Software Engineering: An Agile Unified Methodology, First Edition, Mc Graw-Hill International Edition, 2014.
3. Craig Larman, Applying UML and Patterns, 3rd ed, Pearson Education, 2005.

2304IT552	STARTUP OPPORTUNITIES FOR IT ENGINEERS				L	T	P	C				
					2	0	0	2				
PREREQUISITE:												
	1. Basic understanding of Entrepreneurship											
	2. Problem solving skills											
	3. Business Strategy Management											
	4. Technology fundamentals											
COURSE OBJECTIVES:												
	1. Learn how to harness technology for groundbreaking entrepreneurial ventures.											
	2. Explore creative ways to identify and seize opportunities.											
	3. Learn to gather data and analyze trends to ensure your business concept is viable.											
	4. Develop skills to turn ideas into actionable business propositions and deliver impactful presentations.											
	5. Understand the fundamentals of launching new ventures and building cohesive teams.											
	6. Develop the ability to translate visionary ideas into detailed plans for market execution and financial viability.											
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Explain Concepts of Innovation, Entrepreneurship and Startups in Technology											
CO2:	Develop Startup or Business ideas, minimum viable products and business models for real life problems											
CO3:	Analyze global trends in technology and entrepreneurship to identify opportunities for innovation and competitive advantage.											
CO4:	Apply market research techniques and conduct feasibility analyses to identify opportunities for innovative technological solutions.											
CO5:	Demonstrate teamwork and communication skills in multidisciplinary teams to pitch innovative startup ideas effectively.											
COs Vs POs MAPPING:												
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
	CO1	2	2						1	2	2	3
	CO2	2	3	3	2	2	2		2	2	3	3
	CO3		3		3		2		1	2	2	3
	CO4		3	2	3	2	2		1	2	3	3
	CO5		1						3	3	2	2
COs Vs PSOs MAPPING:												
	COs	PSO1	PSO2	PSO3								
	CO1	1	1	1								
	CO2	1	3	2								
	CO3	1	2	1								
	CO4	1	2	1								
	CO5	1	2	1								
COURSE CONTENTS:												
1.	Introduction to Startups											
2.	Foundations of Entrepreneurship – Strategies & Competencies											
3.	Startup Feasibility Analysis, Survival & Growth											

4.	Ethics and Social Responsibilities in Startups
5.	Startup Success Metrics & Business Plan
6.	Customer – Centric Innovation
7.	Leveraging Technologies for Startup Success
8.	Managing New Venture and Secure Funding
9.	Business Models and Value Proposition
10.	Open Source in Startups
11.	Digital Marketing and Strategies
12.	Minimum Viable Product Development & Tools for Creating MVPs
13.	Patent and Intellectual Property
14.	Effective Team Building
15.	Entrepreneurial Ecosystem Dynamics
TOTAL: 30 HOURS	
Evaluation Procedure	
1.	Assignment 1– Review of IT based Startups – 20 Marks
2.	Assignment 2– Open Source in Startups & Business Models – 30 Marks
3.	Assignment 3 – Review of Business Models of IT based Startups – 20 Marks
4.	Presentation – Idea, Market Research Results, 1st MVP, Possible Fund Raising Model to be adopted, IP – 30 Marks
REFERENCES:	
1.	Zero to One: Notes on Startups, or How to Build the Future by Peter Thiel and Blake Masters
2.	Start with Why: How Great Leaders Inspire Everyone to Take Action by Simon Sinek
3.	The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses by Eric Ries
4.	https://www.edx.org/course/innovation-and-entrepreneurship